

Phosphate plant -- Sensors

Section report: Sensors

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-04-30T23:00:00 (120 days, 2736 operating hours)
Product	Dap from 29.0 % P2O5 ore, 545.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 2880 hourly intervals of 69 signals generated 2026-09-23T15:58:38.796 basis :screening_plant_model

The inferential sensors: process, vision and acoustics

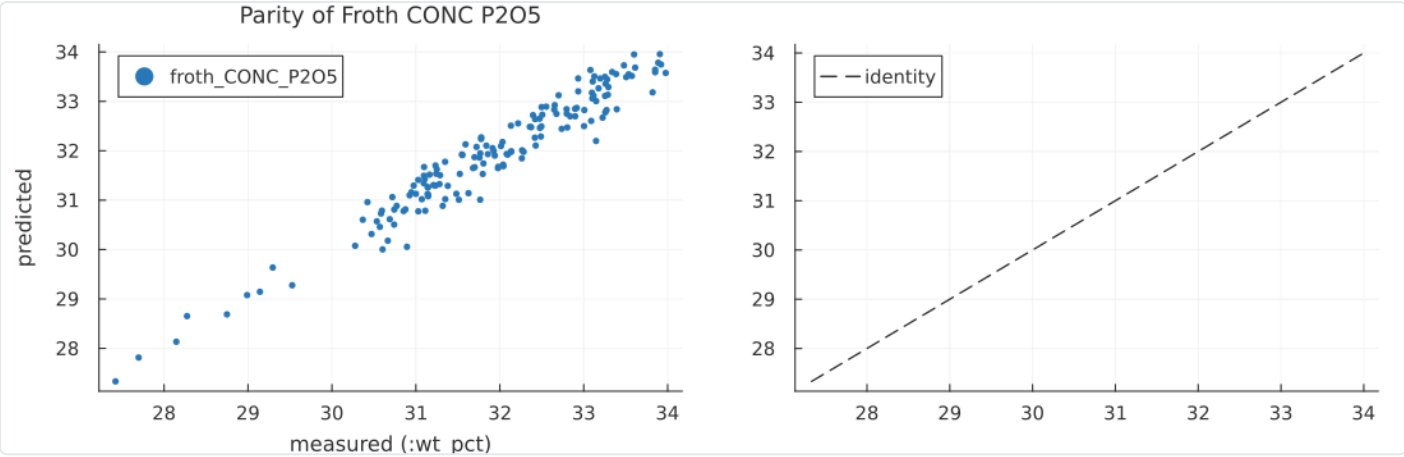
THE REGISTER OF THE INFERENTIAL SENSORS

Quantity	Value									
Sensors	14 inferential models, of which 4 read an image or a sound									
Median R2	0.54428									
Best sensor	:pump_PUMP_VIBRATION									
Status	:at_risk									

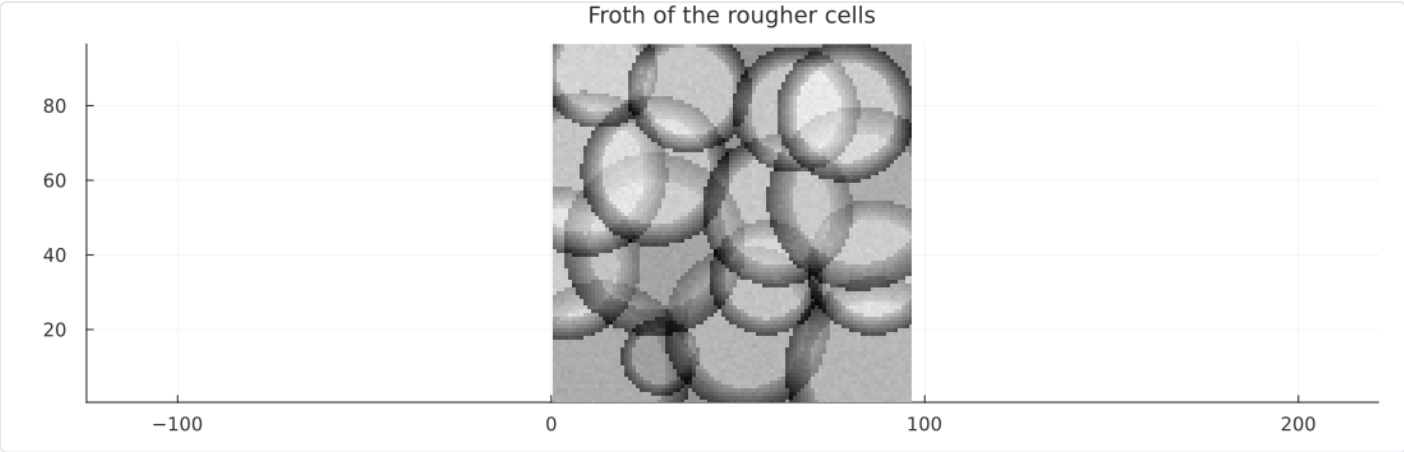
Sensor	Infers	Modality	Unit	R2	RMSE	Bias	In domain %	Drift	Status	Inputs
:pump_PUMP_VIBRATION	:PUMP_VIBRATION	:acoustic	:count	0.99279	0.11026	0.0078	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:froth_CONC_P2O5	:CONC_P2O5	:visual	:wt_pct	0.94903	0.284063	-0.0127	n/a	0	Healthy	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:mill_MILL_POWER	:MILL_POWER	:acoustic	:kw	0.9311	89.0989	-33.98	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:process_STEAM_FLOW	:STEAM_FLOW	:process	:t_ph	0.91184	1.9597	0.0542	100	0.0217	Healthy	EVAP_FEED_FLOW, STRONG_ACID_P2O5, EVAP_VACUUM
:process_CONC_P2O5	:CONC_P2O5	:process	:wt_pct	0.89048	0.456288	0.0048	100	0.0019	Healthy	FLOT_FEED_P2O5, CYCLONE_P80, REAGENT_FLOW
:process_MILL_POWER	:MILL_POWER	:process	:kw	0.80293	202.537	-11.22	100	-4.49	Healthy	ROM_FEED, SLURRY_DENSITY, MILL_FEED_WATER
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:process	:wt_pct	0.58554	0.203578	-0.02	100	-0.008	Healthy	SLURRY_SG, ATTACK_TEMP, FREE_SO4, WASH_WATER_FLOW
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:process	:wt_pct	0.50303	0.316116	0.01	100	0.004	Healthy	EVAP_TEMP, STEAM_FLOW, EVAP_VACUUM
:process_CYCLONE_P80	:CYCLONE_P80	:process	:um	0.44705	9.5032	0.592	100	0.2368	Healthy	MILL_POWER, SLURRY_DENSITY, CYCLONE_PRESSURE, MILL_FEED_WATER
:process_PRODUCT_N	:PRODUCT_N	:process	:wt_pct	0.27181	0.292159	-0.0164	100	-0.0066	Healthy	AMMONIA_FLOW, PRODUCT_FLOW, MELT_DENSITY
:process_FILTER_RATE	:FILTER_RATE	:process	:t_per_m2_d	0.23443	1.1356	0.0196	100	0.0079	Healthy	FILTER_VACUUM, WEAK_ACID_P2O5, SLURRY_SG, GYPSUM_FREE_P2O5
:rock_CYCLONE_P80	:CYCLONE_P80	:visual	:um	0.17152	9.0556	0.4531	n/a	0	Degraded	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:process	:wt_pct	0.05478	0.024921	-0.0003	100	-0.0001	Healthy	FILTER_RATE, FILTER_VACUUM, WASH_WATER_FLOW, WEAK_ACID_P2O5
:process_ATTACK_TEMP	:ATTACK_TEMP	:process	:deg_c	-0.00493	16.2509	-0.0224	100	-0.009	Healthy	H2SO4_FLOW, ROCK_FEED, RECYCLE_ACID_FLOW, SLURRY_SG

WHAT THE SENSORS SAY AGAINST WHAT THE PLANT MEASURED

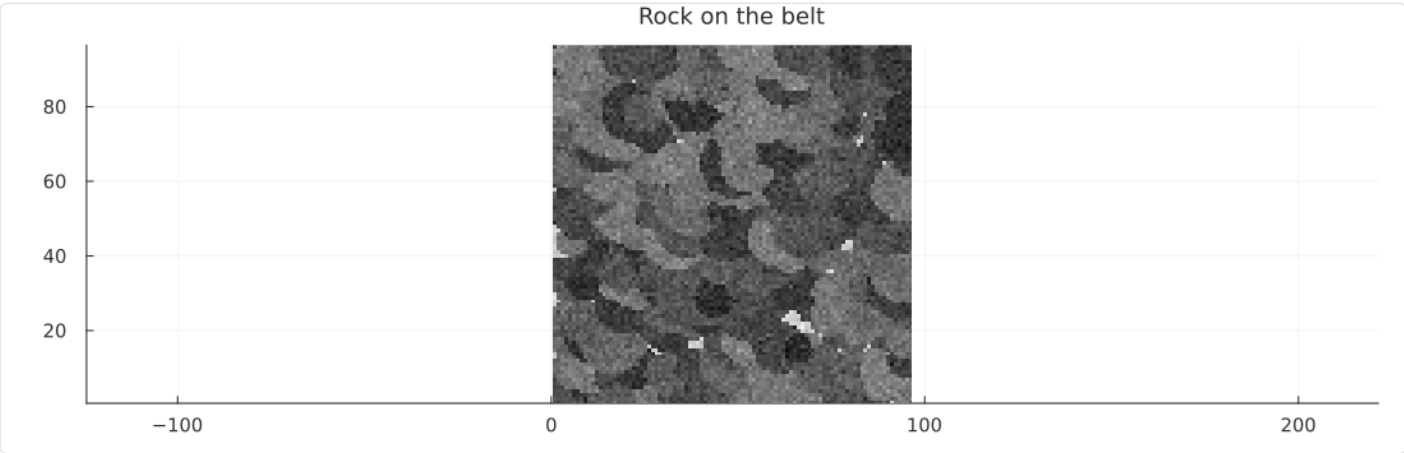
Sensor	Infers	Unit	Estimate	Measured	Difference %	RMSE	Status
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:wt_pct	55.74	55.89	-0.269	0.3161	Compliant
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:wt_pct	28.77	28.72	0.1929	0.2036	Compliant
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:wt_pct	0.5012	0.5101	-1.76	0.0249	Compliant
:process_CYCLONE_P80	:CYCLONE_P80	:um	151.21	148.3	1.96	9.5	Compliant
:process_MILL_POWER	:MILL_POWER	:kw	5,453.54	5,489.39	-0.653	202.54	Compliant
:process_PRODUCT_N	:PRODUCT_N	:wt_pct	18.14	18.2	-0.327	0.2922	Compliant
:process_FILTER_RATE	:FILTER_RATE	:t_per_m2_d	8.54	8.92	-4.18	1.14	Compliant
:process_CONC_P2O5	:CONC_P2O5	:wt_pct	32.85	32.95	-0.2797	0.4563	Compliant
:process_ATTACK_TEMP	:ATTACK_TEMP	:deg_c	77.88	77.37	0.6672	16.25	Compliant
:process_STEAM_FLOW	:STEAM_FLOW	:t_ph	52.22	52.41	-0.3654	1.96	Compliant



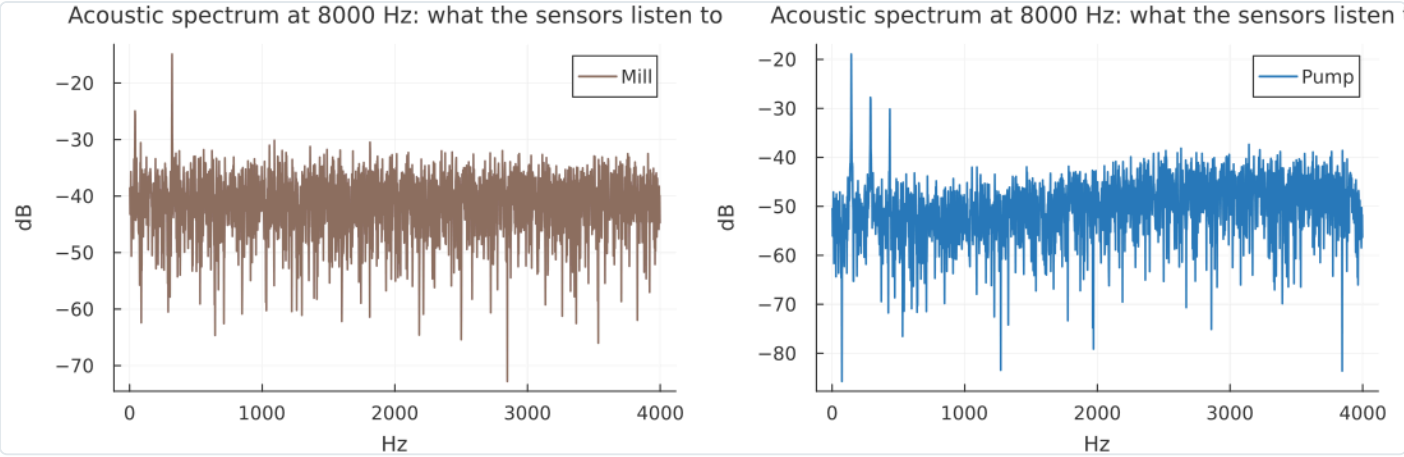
Parity of the froth sensor against the analyser



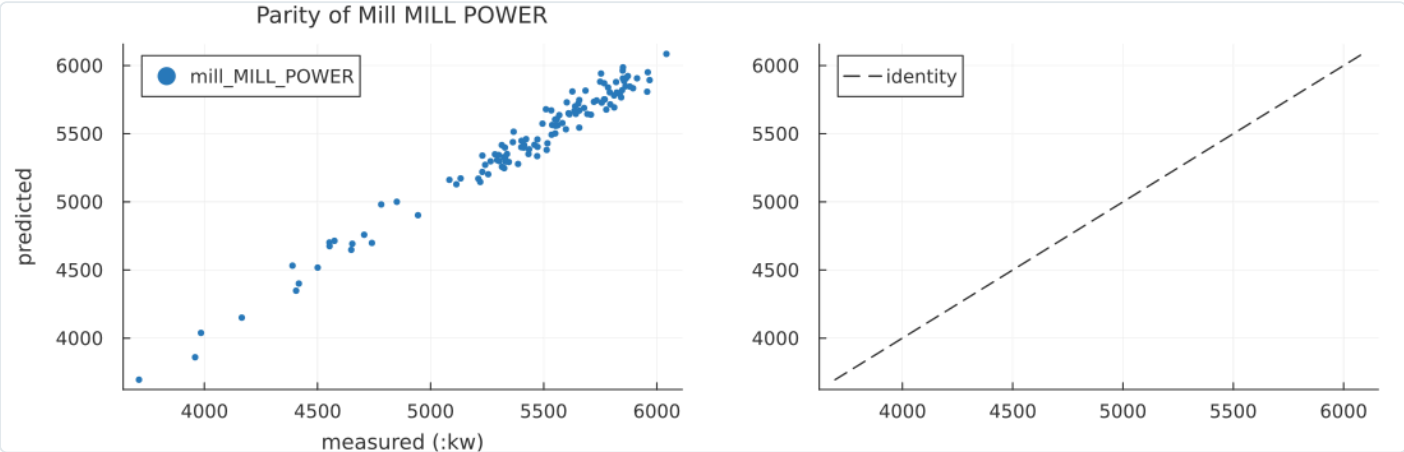
The froth image the visual sensor reads the grade from



The image of the rock the visual sensor reads the grind from



Spectra of the mill and of the pump: the two acoustic channels of the plant



Parity of the acoustic sensor of the mill against the power it draws